What multi-omic network inference captures and misses: CORNETO on a patient stem-cell model of progressive multiple sclerosis

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October 13, 2026

1 Introduction

Network biology treats disease as a perturbation that spreads through networks of interacting molecules [1]. A common way to study it is to start from a large prior-knowledge network (PKN) of literature-curated, signed and directed interactions and to ask which part of it explains the changes measured in one disease model. This step, called contextualisation, turns a generic map into a cell-specific hypothesis [2]. CARNIVAL [3] and COSMOS [4] do this for transcription factor (TF) activities and metabolites, and CORNETO [5] unifies them. Each of these methods searches for the smallest sub-network that connects some starting nodes to the measured nodes with the observed signs (Fig. 1a,b).

The result is a hypothesis, not a measurement, and on patient data there is no ground truth to check it against. What the method returns depends on the network as much as on the data:

- which edges the PKN contains;
- which nodes may start a signal;
- how strongly small networks are preferred.

How much of an inferred network is driven by the data, and how much by the structure of the PKN, is rarely tested.

We use a published hypothesis as a test case. In induced neural stem cells (iNSCs) from patients with progressive multiple sclerosis (PMS), Ionescu et al. found a senescent, neurotoxic state that a cholesterol-synthesis inhibitor, simvastatin, reduced [6]. They proposed a route from HMG-CoA reductase (HMGCR) through cholesterol to cholesterol-responsive TFs (E2F1, EGR1, WT1, SP1, JUN, JUNB). We asked three network questions (Fig. 1c):

1. Is the proposed route present in the PKN?
2. Does contextualisation with CORNETO select it, when it is not given as a starting node?
3. How much of the selected network comes from the data, and how much from the network?

2 Methods

2.1 Data and measured nodes

We used bulk RNA-seq of 3 control and 3 PMS iNSC lines (GSE297192 [7, 8]) and metabolomics and lipidomics of 3 control and 4 PMS lines (Metabolomics Workbench ST003331/ST003332/ST003328 [6, 9]). Each cell line is one observation, and replicates were averaged. Gene expression differences between PMS and control lines were scored with an empirical-Bayes moderated t statistic [10]. TF activities were inferred from these scores with CollecTRI regulons and decoupler [11, 12]. The 50 most changed TFs, all of which kept their sign when any one line was left out, became the measured nodes. In the joint analysis, 36 metabolite contrasts mapped to the PKN were added, including free cholesterol.

2.2 Networks

The signalling PKN is OmniPath [13]: signed, directed and curated edges between expressed genes (9,870 edges, 3,443 nodes). The multi-omic PKN is COSMOS [4]. It adds metabolic reactions from Recon3D [14] and compound–protein interactions [15], with metabolites as nodes (54,571 edges, 26,510 nodes).

2.3 Contextualisation

CORNETO's CarnivalFlow method [5] selects the subnetwork that best reproduces the signs of the measured nodes. A penalty λ on the number of edges makes it prefer small networks. Any PKN node without incoming edges could act as a starting node, and the method chose its sign. The proposed route (HMGCR, SREBF1/2, SCAP, INSIG1, the six TFs, mevalonate, cholesterol) was never allowed to start a signal, so finding it would not be circular. We fitted TFs alone on each PKN and TFs plus metabolites on COSMOS. For each fit, we chose the largest λ that kept the fitting error within 10% of the unpenalised fit, a rule set before any network was inspected. The fits were solved as mixed-integer programs with HiGHS [16]. Settings: CORNETO 1.0.0rc8; $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$; 1% optimality gap; 30 min (OmniPath) and 60 min (COSMOS) time limits. Code, settings and checksums of every input are recorded in the project repository.

2.4 Route analysis and controls

For question 1, we listed every reaction of the 20 cholesterol-synthesis enzymes (KEGG [17]). We checked whether the pathway intermediates can be reached in order, and searched for the shortest HMGCR → cholesterol path in the metabolic part of COSMOS. That search excluded ubiquitous co-factors and penalised hub metabolites [18]. Missing reactions were added to a repaired PKN. For question 2, we also gave CORNETO a known perturbation: HMGCR inhibited, with the simvastatin-induced fall in free cholesterol as the only measurement. For question 3, we repeated the OmniPath fit:

- on ten randomised PKNs that keep every node's degree and the edge signs;
- under all 20 assignments of the six lines to two groups;
- without each line in turn;
- with starting nodes restricted to cell-surface receptors.

We also compared the networks selected across λ values and PKNs. Overlaps are Jaccard indices on edges and on inferred (unmeasured) nodes.

3 Results

Measured nodes. SP1 and JUN were the second and sixth most activated of 628 TFs in PMS lines, and the AP-1 and NF-κB complexes ranked 13th and 25th (Fig. 2a). JUNB and EGR1 were moderately increased, E2F1 was unchanged, and WT1 is not expressed. The senescence markers SERPINE1 (up) and LMNB1 (down) changed, but sterol-synthesis genes such as HMGCR were not induced (Fig. 2b).

3.1 The route is in the network, but not uniquely

The cholesterol pathway is almost complete in COSMOS: 56 of its 58 reactions are present. The two missing reactions are halves of the squalene-synthase step (FDFT1), and they disconnect the pathway between farnesyl and squalene (Fig. 3a, row 1). After adding them, the shortest weighted path from HMGCR to cholesterol followed the textbook pathway. Cholesterol is two edges away from SP1, JUN and E2F1. But 33–66% of degree-preserving randomised networks connect it to these TFs just as closely, so short paths alone do not support the route.

3.2 Inference does not select it, unless told where to start

Neither PMS fit used the pathway (Fig. 3a, rows 4–5). HMGCR stayed inactive, and the measured rise in cholesterol was explained by a single transport reaction (Fig. 3b). When CORNETO was told that HMGCR is inhibited and that cholesterol falls, it recovered the pathway through mevalonate, using 16 of the 20 enzymes in order, but only in the repaired network. In the original network it reached cholesterol by a 32-edge detour through unrelated mitochondrial reactions (row 2). That detour is sign-consistent but biochemically meaningless.

Adding metabolites changed the routes more than the nodes. The three networks shared 50–71% of their protein nodes but only 7–21% of their edges. The kinases MAPK1, MAPK14, PRKACA, FYN and AKT1 were present in all of them. In the joint network, every route into SP1, JUN, JUNB and EGR1 started from one unmeasured node, a tamoxifen compound, which alone sat upstream of 39 of the 79 explained measurements (Fig. 4).

3.3 Much of the network comes from the network

- **One hub, arbitrary identity.** At every sparsity level, one starting node sat upstream of 24–92% of the explained measurements. Which node this hub was changed with λ and with the PKN (Fig. 5a). Only 6–24% of edges survived across all five λ values (Fig. 5b).
- **Random networks fit too.** Six of ten degree-preserving random networks explained every TF exactly, while sharing only 7–18% of the inferred nodes with the real network. A perfect fit is therefore no evidence for the topology.
- **The grouping, not the network, carries the signal.** Only the true split of the lines (and its mirror) was fitted exactly; the 18 other splits were not ($p = 2/20$). Still, the mirror problem, which differs only in sign, shared just 47% of its edges with the true network. Leaving out one line changed 41–65% of the inferred nodes.

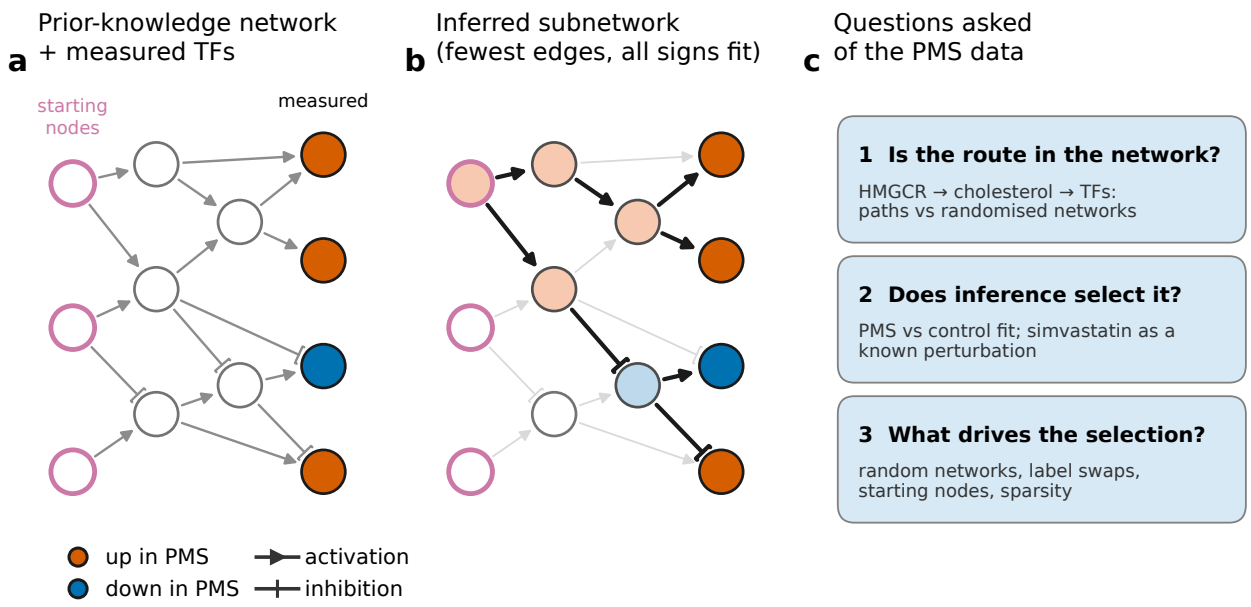


Figure 1: Network contextualisation and the questions asked. (a) A prior-knowledge network (PKN) of signed, directed interactions, with TFs measured as up or down in PMS. Nodes without incoming edges may start a signal. (b) CORNETO selects the subnetwork with the fewest edges whose signal reproduces every measured sign; light colours are inferred states. (c) The three questions asked of the PMS iNSC data. Toy example; the real PKNs have 9,870 (OmniPath) and 54,571 (COSMOS) edges.

- **Starting nodes decide the shape.** Restricting the starting nodes to receptors gave the same picture: a single receptor (PTPRE) explained 49 of 50 TFs. Removing the 3,246 compound starting nodes from the joint fit still fitted every TF, at a slightly higher error (9.4 vs 8.9). It replaced tamoxifen by another single hub, the extracellular-matrix protein LAMA5 (33 of 79 measurements). Only then did a route run from cholesterol to a hypothesis TF (cholesterol → MAPK3 → ⋯ → JUNB). Without any metabolite starting nodes, JUN and EGR1 were also reached through cholesterol. In both cases cholesterol was still supplied by transport, not synthesis.
- **The optimiser is not exact.** Some networks the solver reported as optimal scored up to 19% worse than other known solutions (Fig. 5c). A fit of all six lines at once found no solution with $\lambda > 0$.

4 Discussion

From a network-biology perspective, the main lesson is that contextualisation reflects the structure of the PKN at least as much as the data. Two parts of the result were robust:

- **A local kinase layer.** MAPK1, MAPK14, PRKACA and FYN, directly upstream of the activated AP-1 and SP1 targets, appeared across settings, networks and data types.
- **The pathway under a known perturbation.** With the perturbation stated, CORNETO recovered the textbook cholesterol pathway, once two missing edges were restored. A gap that cannot be seen in the network itself is enough to hide a well-known pathway.

The upstream part of the network, where the proposed mechanism would sit, was not identifiable.

1. **Too much freedom at the start.** When thousands of nodes may start a signal, a sparse solution collapses onto a single hub, and many hubs are equally good. In COSMOS most candidate starting nodes (3,246 of 4,711) are drug and food compounds rather than cellular signals, which is how tamoxifen and cinnamaldehyde became hubs. Removing them moved the hub to a protein instead of removing it. Random networks fitting as well as the real one makes the same point: the data constrain the last step into the TFs, not the path before it.
2. **Signs are not chemistry.** COSMOS encodes metabolism as signed edges without mass balance, so transport or an unrelated reaction can explain a metabolite as well as its synthesis. This is

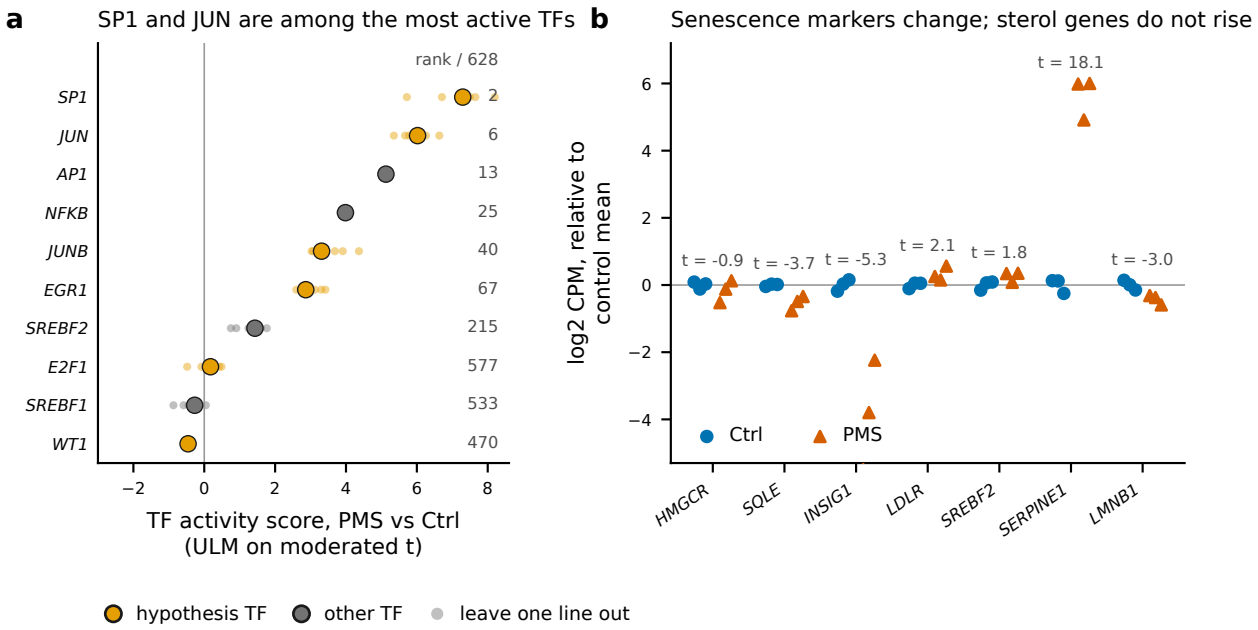


Figure 2: The measured nodes: TF activity and cholesterol-related genes, PMS versus control (3 vs 3 lines). (a) TF activity scores for the six hypothesis TFs (orange), the SREBFs and the AP-1/NF-κB complexes. Large dots: all lines; small dots: leaving out one line; numbers: rank among 628 TFs. (b) Per-line expression relative to the control mean, with the moderated t statistic. Senescence markers change, while sterol-synthesis genes do not rise.

how the transport shortcut and the mitochondrial detour arise. That the PMS lines show no transcriptional induction of sterol synthesis makes the shortcut easier still.

3. **Optimisation is approximate.** Solver guarantees were unreliable on the large network, so model selection based on fitting error inherits solver noise.

These conclusions rest on 3 control and 3–4 PMS lines, with data layers from different experiments that are not matched line by line. Disease status is also confounded with donor background. The positive control and the starting-node restrictions were added after the main fits. Limiting starting nodes to measured perturbations and using stoichiometric metabolic models are the clearest next steps. Removing compound nodes alone was not enough. Comparing single-modality networks and factor models (MOFA/SOFA [19, 20]) remains future work.

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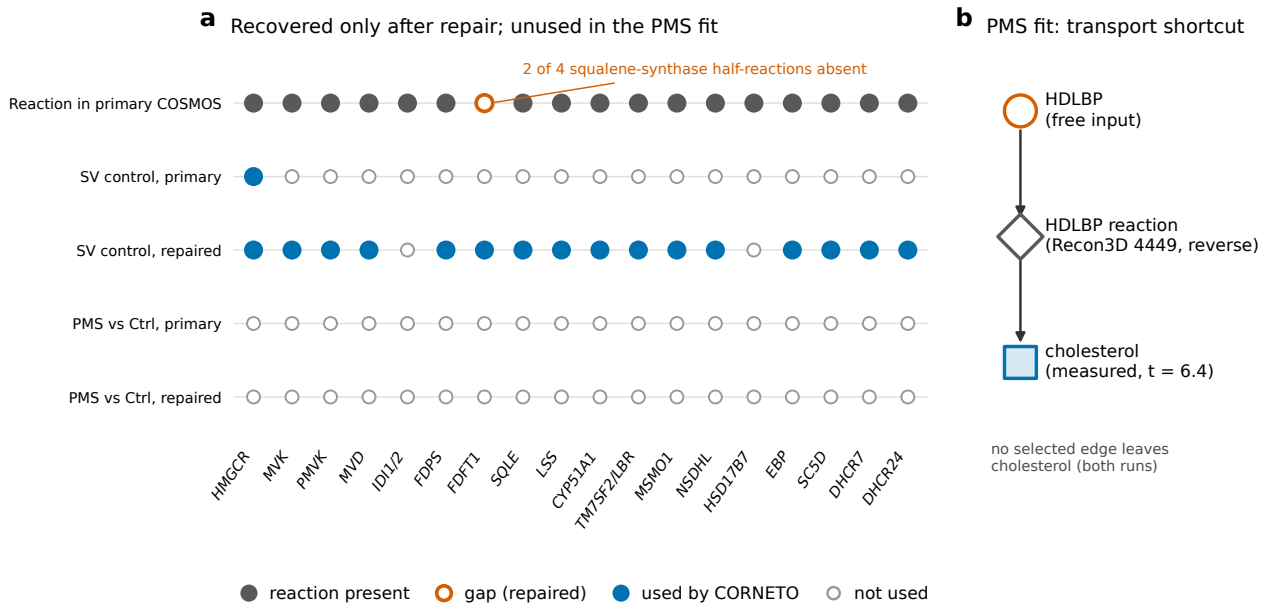


Figure 3: The cholesterol route is in the network, but inference uses it only for a known perturbation and only after repair. (a) The cholesterol-synthesis enzymes in pathway order. Row 1: present in COSMOS; two squalene-synthase (FDFT1) half-reactions are missing and were added in the repaired network. Rows 2–3: enzymes CORNETO used when told that HMGCR is inhibited and cholesterol falls (simvastatin), in the original and the repaired metabolic network. Rows 4–5: enzymes used in the PMS-versus-control fits. (b) In both PMS fits, the cholesterol rise is explained by one transport reaction.

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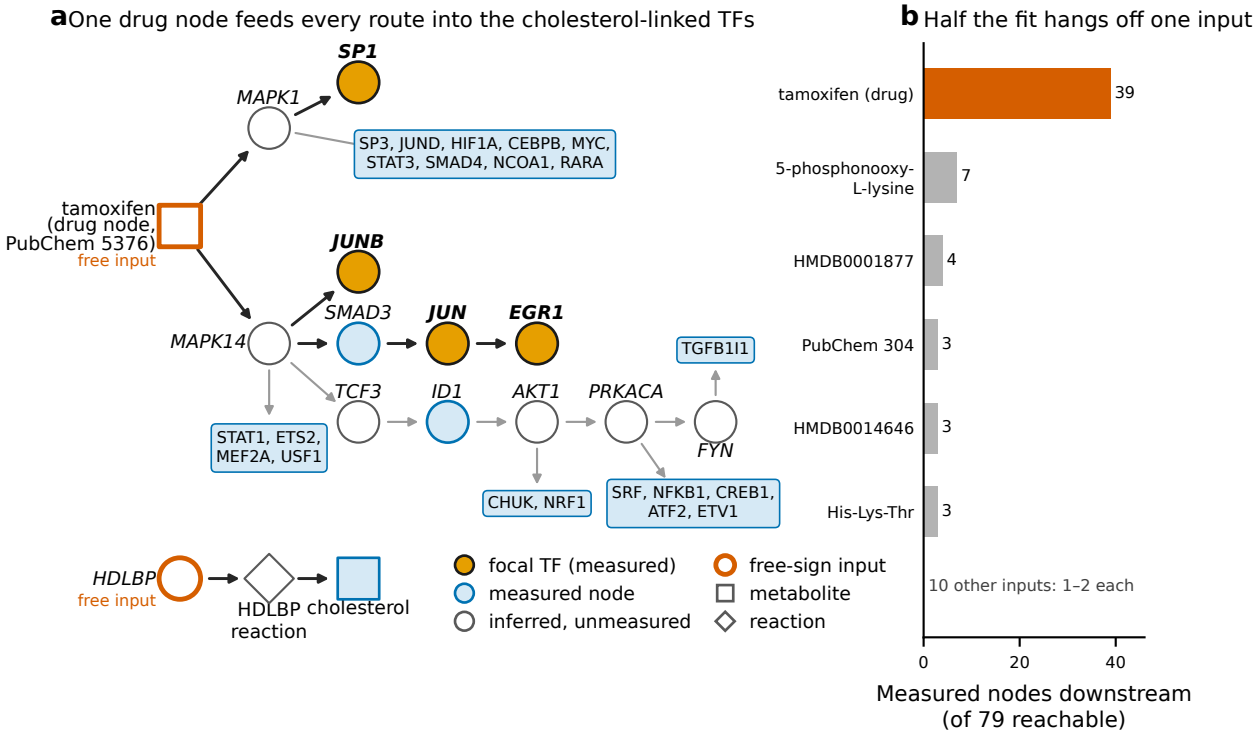


Figure 4: A single compound hub feeds the joint network. The TF-plus-metabolite network on COSMOS ($\lambda = 0.1$), restricted to the routes into the cholesterol-linked TFs. (a) All routes into SP1, JUNB, JUN and EGR1 start at one unmeasured starting node, a tamoxifen compound, and pass through the kinases MAPK1 or MAPK14. Blue boxes collapse further measured TFs. All shown nodes are inferred as increased in PMS. (b) Measured nodes downstream of each starting node used.

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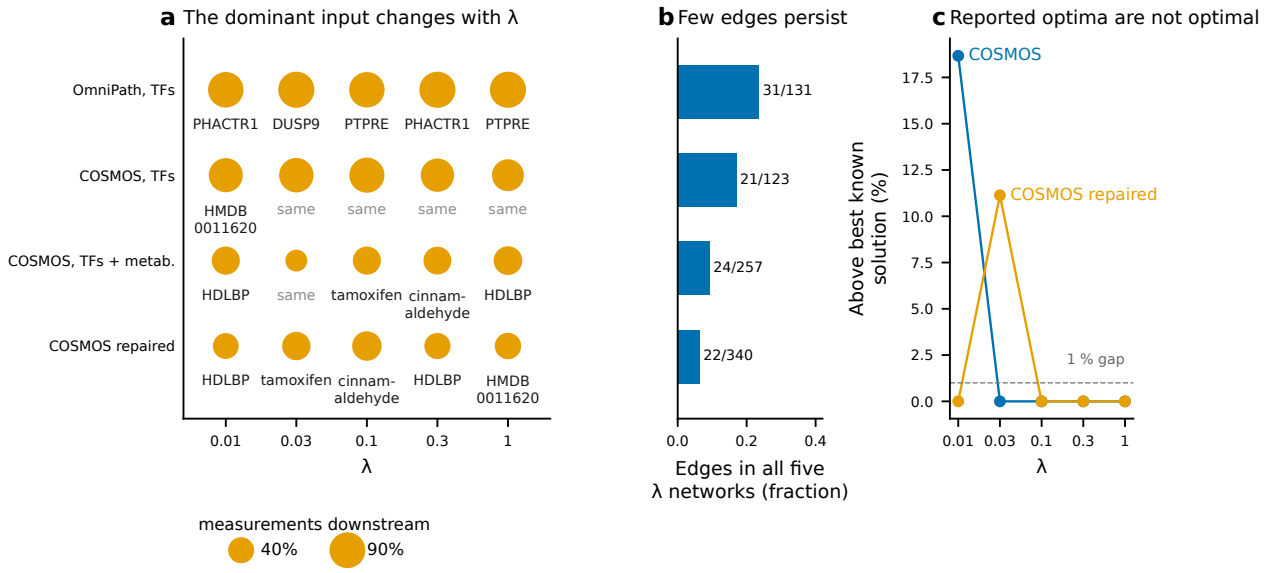


Figure 5: What changes and what persists across sparsity settings. (a) For each fit and penalty λ, the starting node upstream of the most measurements (dot area: fraction of measurements; "same": unchanged from the previous λ). (b) Edges selected at all five λ values, as a fraction of all edges ever selected. (c) Each saved network was scored under every λ; the value shown is how much worse the network the solver returned as optimal is than the best known network.

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